STOCK MARKET PRICE PREDICTION USING MACHINE LEARNING

Proposal Presentation

GROUP 2025-Y2-S1-MLB-B6G2-01

This presentation outlines the plan for building an ML model to predict stock prices using historical market data and Python.

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INTRODUCTION

The stock market is a complex system influenced by numerous factors such as economics, politics, and investor sentiment.

Predicting stock prices accurately is crucial for traders and investors to make informed decisions.

Machine Learning leverages historical data patterns to create prediction models that can support strategic investment planning.

PROBLEM DOMAIN

Domain: Finance & Investment Analytics.

Real-World Problem: Investors often rely on manual analysis, which is time-consuming and prone to error.

A data-driven ML approach can automate analysis and detect subtle patterns in market data.

Impact: Helps reduce financial risk, improve profitability, and provide quick decision-making insights.

PROJECT OBJECTIVES

1. Collect and preprocess historical stock data from reliable sources such as Yahoo Finance.

2. Engineer features like moving averages, RSI, and MACD to improve prediction performance.

3. Experiment with different machine learning models such as Linear Regression, Random Forest, and LSTM.

4. Evaluate models with metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2.

5. Deploy the final model in a web-based dashboard for easy visualization and interaction.

DATASET SELECTION

Source: Yahoo Finance / Alpha Vantage APIs.

Type: Time-series financial data.

Features include Date, Open, High, Low, Close, Volume, plus engineered indicators.

Dataset contains over 1,000 daily records for each selected stock, meeting assignment requirements.

Reliable, up-to-date, and consistent data makes it suitable for ML-based prediction tasks.

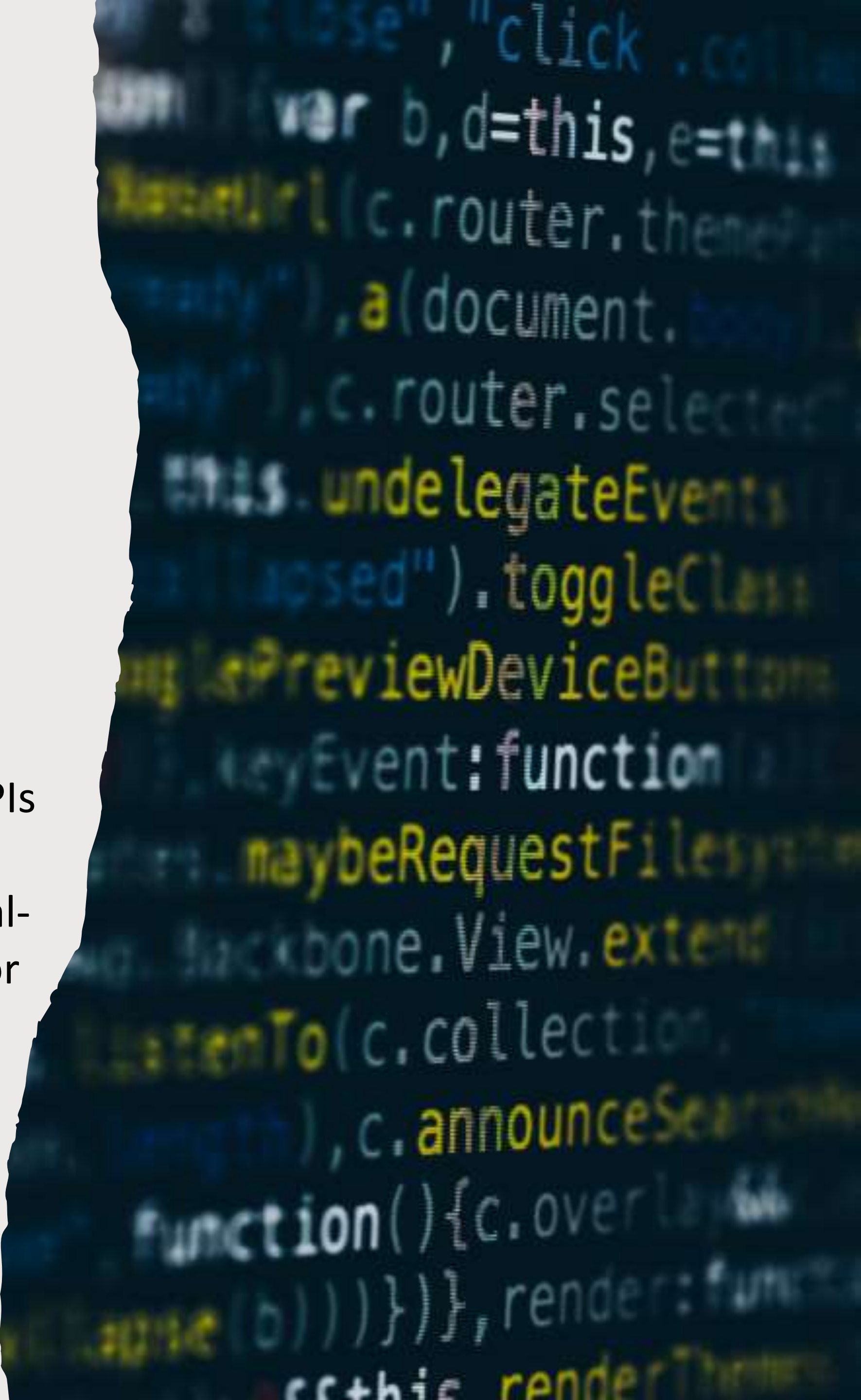
DATASET JUSTIFICATION

Contains a variety of features essential for price prediction, including price levels, trading volume, and time-based patterns.

Large enough to train models without overfitting and small enough for efficient computation.

Data is clean and requires minimal preprocessing compared to alternative datasets.

Availability of APIs ensures the possibility of real-time updates for future development.



KEY PREDICTIVE FEATURES

Basic features: Open, High, Low, Close, Volume.

Engineered features:

- Moving Averages (MA5, MA20) for trend detection.

- Relative Strength Index (RSI) for market momentum.

- MACD for identifying price reversals.

- Bollinger Bands for volatility analysis.

Target Variable: Closing Price, as it reflects investor sentiment at the end of a trading day.

PROPOSED METHODOLOGY

Step 1: Data Collection – Use APIs to gather historical price data.

Step 2: Preprocessing – Handle missing data, normalize values, and split into training/testing sets.

Step 3: Feature Engineering – Calculate indicators such as MA, RSI, MACD.

Step 4: Model Development – Compare multiple algorithms to identify the best performer.

Step 5: Evaluation – Use MSE, RMSE, and accuracy metrics to assess performance.

Step 6: Deployment – Build an interactive dashboard to display predictions.

EXPECTED OUTCOMES

A trained ML model that can predict short-term stock prices with reasonable accuracy.

Comparison of model performances to choose the most reliable one.

A functional dashboard for visualization of stock trends and predictions.

A reusable methodology that can be adapted to other financial forecasting problems.

LIMITATIONS & ETHICAL CONSIDERATIONS

Limitations:

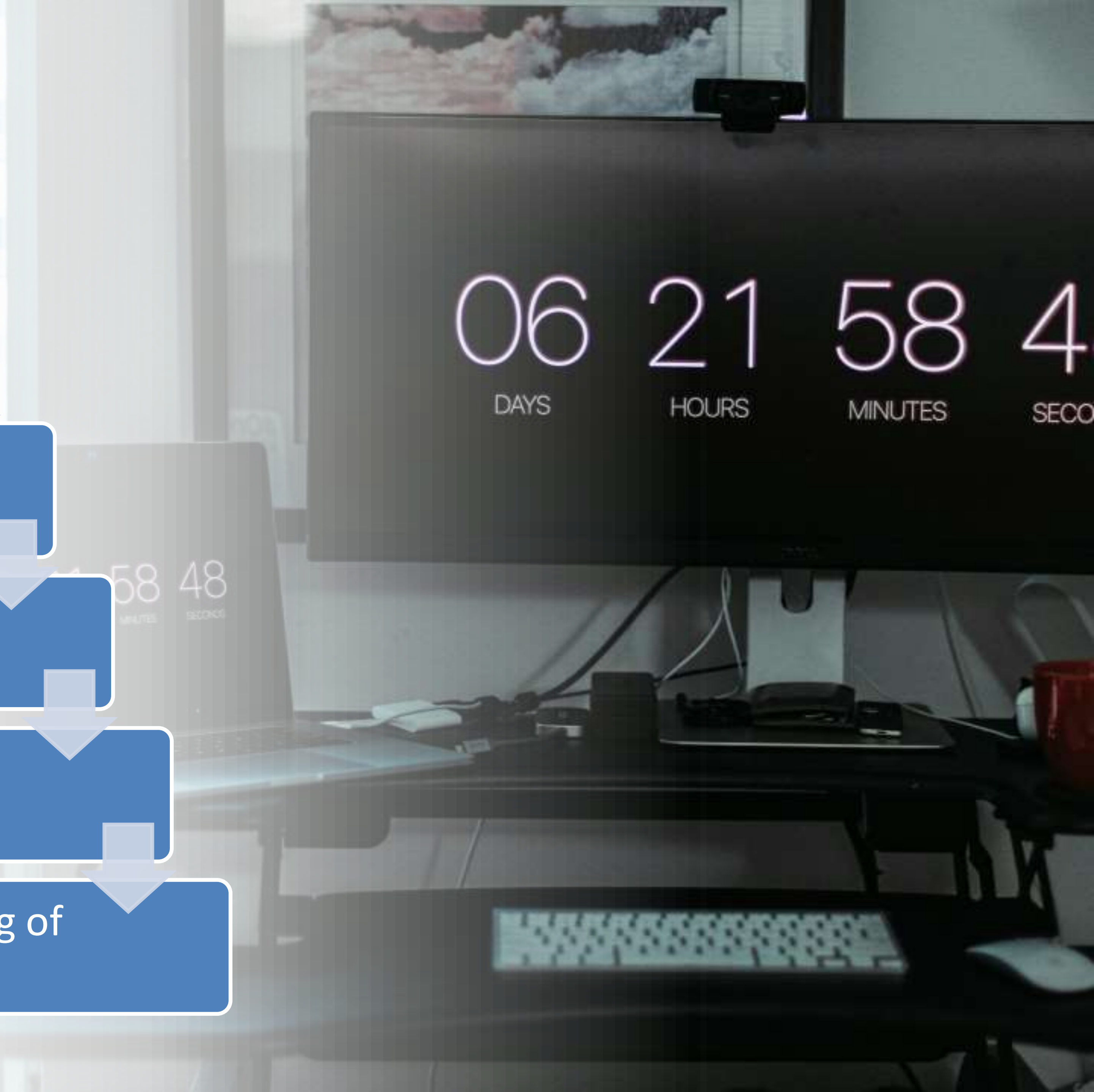
- Stock market prices can be influenced by unpredictable global events.
- Potential overfitting if the dataset is not diverse enough.

Ethical Considerations:

- Predictions should be treated as advisory and not absolute guarantees.
- Avoid using predictions to manipulate the market.
- Be transparent about the model’s accuracy and limitations.

PROJECT TIMELINE

- Week 1–2: Data collection & preprocessing.
- Week 3–4: Feature engineering & exploratory data analysis.
- Week 5–6: Model training & evaluation.
- Week 7: Model optimization based on test results.
- Week 8: Deployment & testing of the dashboard.



CONCLUSION

Machine Learning with Python provides a powerful way to forecast stock prices.

Our project will combine technical indicators and ML models for improved accuracy.

Final deployment will be in an interactive dashboard format.

Future improvements include integrating real-time news sentiment and deep learning models like LSTM.

we look forward to your thoughts.